

Blockchain-Assisted Mission Trust in Air–Surface–Underwater Unmanned Systems: A Cross-Layer Review and Object–Assurance–Lifecycle Framework

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Abstract

Air–surface–underwater missions combine unmanned aerial vehicles, unmanned surface vehicles, and unmanned underwater vehicles across heterogeneous links, intermittent connectivity, and multiple administrative domains. Security studies usually examine one platform, network, or ledger mechanism at a time, leaving a gap between trustworthy observations and defensible mission decisions. This narrative review synthesizes literature on cross-domain cooperation, unmanned-system security, trust and reputation, provenance, remote attestation, zero trust, and permissioned blockchains. It proposes the Object–Assurance–Lifecycle (OAL) framework, which relates four trust objects (entity, platform/device, link/path, and data product) to four assurance layers (observation validity, commitment and provenance integrity, ledger and governance consistency, and mission-use suitability) across six mission stages. An object-dependency graph and a context–temporal–graph reasoning loop explain how evidence and attacks propagate without treating missing or degraded evidence as proof of malice. The review also defines a strict blockchain boundary: distributed ledgers can coordinate membership, commitments, versions, audit, and post-partition reconciliation, but cannot establish physical truth, secure compromised endpoints, remove partitions, guarantee hard real-time control, or replace governance. An evidence-maturity ladder and a proposed Cross-Domain Unmanned Trust Profile turn the synthesis into a testable research and standardization agenda.

Keywords: mission trust; unmanned systems; UAV–USV–UUV cooperation; blockchain; data provenance; remote attestation; zero trust; maritime autonomy

1. Introduction

Unmanned maritime operations are becoming systems of systems. A UAV can rapidly survey a wide area and maintain a radio link; a USV can loiter, carry compute and communications equipment, and bridge air and underwater segments; a UUV can collect close-range observations where radio and satellite services do not reach. Cross-domain control studies increasingly coordinate such platforms for search, tracking, formation, navigation, and surveillance missions [1–7]. The complementary capabilities are operationally attractive, but they also couple domains whose communication physics, failure modes, owners, and security assumptions differ substantially. Broad reviews of UAV networks and applications emphasize mobility, limited energy, intermittent links, and safety constraints [8–10]; reviews of unmanned marine and underwater vehicles add acoustic delay, difficult localization, harsh deployment conditions, and constrained recovery [11–13]. A mission

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decision may therefore depend on evidence that has crossed several media, gateways, processing stages, and administrative boundaries.

This creates a *mission-trust gap*. Conventional cybersecurity questions—who authenticated, which packet arrived, whether a file hash matches, or whether replicas agree—remain necessary, yet none alone answers whether a particular data product is suitable for a particular decision at a particular time. A correctly signed sonar report can be physically implausible; a valid observation can be replayed after it becomes stale; a correctly ordered ledger entry can originate from a captured endpoint; and a well-provenanced product can still be unsuitable for a high-consequence task because its uncertainty is too large. Research on UAV and swarm security catalogs spoofing, jamming, routing attacks, capture, malware, data manipulation, and privacy threats [14–17]. Underwater-network security reviews describe related attacks under low bandwidth, long propagation delay, and expensive retransmission [18–21]. These literatures identify important mechanisms, but the unit of analysis is often a platform or network rather than the chain from physical observation to mission use.

Blockchain is frequently proposed as an answer to distributed trust. UAV-oriented reviews report uses in identity, authentication, coordination, access control, data sharing, and audit [22–26]. Representative proposals use ledgers for swarm identity, lightweight authentication, trusted message exchange, and distributed data acquisition [27–31]. Maritime reviews similarly associate blockchain with documentation, information sharing, authentication, and inter-organizational coordination [32–34]. The attraction is understandable: several organizations can maintain a shared, append-oriented record without granting one participant unilateral control over history. However, the word *trust* becomes ambiguous when replicated-state consistency is allowed to stand for sensor truth, endpoint integrity, privacy, or timeliness. General blockchain security and Internet-of-Things (IoT) surveys repeatedly document scalability, resource, governance, privacy, and interoperability constraints [35–38]. A useful review must therefore say not only where a ledger helps, but what it does not establish and when it is unnecessary.

Adjacent research offers parts of the answer. Zero trust rejects implicit confidence based only on network location and instead emphasizes explicit, least-privilege, continuously evaluated access [39,40]. Remote attestation separates the production of device evidence, its appraisal, and the relying party’s decision [41–43]. Provenance models describe how entities, activities, and agents are related and can support accountability across transformations [44–47]. Trust-management research supplies reputation, probabilistic, subjective-logic, fuzzy, and learning-based methods [48–53]. Yet these mechanisms are normally discussed in separate architectural vocabularies. A cross-domain mission requires them to be aligned around the objects being judged, the assurance claim being made, and the stage at which evidence is needed.

This review advances one central argument: *mission trust is a chain of non-substitutable assurance claims, and blockchain can strengthen selected transitions in that chain but cannot complete the chain by itself*. To make the argument operational, we propose the authors’ *Object–Assurance–Lifecycle* (OAL) framework. It relates four trust objects—entity, platform/device, link/path, and data product—to four assurance layers—observation validity, commitment and provenance integrity, ledger and governance consistency, and mission-use suitability—over six lifecycle stages from admission to recovery. A cross-cutting assurance and governance plane carries identities, attestation results, provenance commitments, trust state, policy and model versions, ledger state, and consortium rules. It is a plane of mechanisms, not a fifth mission object.

The review makes five further contributions. First, it represents dependencies through a typed object graph with relations such as *controls*, *hosts*, *observes*, *transmits*, *transforms*,

endorses, records, and consumes. This graph joins evidence propagation to attack propagation. Second, it reframes dynamic trust as a context-temporal-graph (CTG) reasoning loop rather than as a new calibrated algorithm: evidence is appraised, attributed, propagated with uncertainty, converted into a policy action, audited, and fed back. Third, it introduces a unified threat-analysis sentence—attacker, capability, object, propagation path, mission impact, and control—while explicitly separating malice, benign environmental degradation, and insufficient evidence. Fourth, it supplies an evidence-maturity ladder from conceptual argument to sustained deployment. Fifth, it proposes a future Cross-Domain Unmanned Trust Profile (CDUTP) that maps mission-trust evidence to existing zero-trust, attestation, provenance, communications, and permissioned-ledger specifications. The profile is an author proposal, not a current standard.

The scope is deliberately public and defensive. Civil applications and publicly documented military-support settings are included because coalition ownership, contested communication, and mission criticality sharpen the governance problem. The review does not discuss weapon effects, targeting procedures, or sensitive tactics. It reports no new experiment, dataset, performance result, protocol proof, or calibrated trust score. Its originality lies in conceptual integration, boundary setting, and an evaluation agenda that can be tested by future work.

The remainder of the paper describes the review scope and analytical lens (Section 2), establishes the operational and conceptual foundations, compares fragmented research landscapes, develops OAL and CTG reasoning, examines blockchain patterns and boundaries, organizes evaluation evidence by maturity, and closes with research propositions, limitations, and conclusions.

2. Review Scope and Analytical Lens

This article is a narrative, problem-driven synthesis rather than an exhaustive inventory. The principal publication window is 2016 to 6 August 2026, with earlier foundational work retained where it defines reputation, distributed-system limits, provenance, or consensus concepts. Only English-language evidence is used. Priority is given to peer-reviewed original studies, authoritative reviews, standards, and official technical specifications; preprints are suitable only for clearly identified frontier developments and do not independently support central claims. Searches were organized around six intersecting themes: UAV blockchain and swarm security; maritime and underwater communication and security; heterogeneous UAV-USV-UUV cooperation; IoT trust and reputation; provenance, attestation, and zero trust; and permissioned-ledger architecture and governance. Seed reviews were extended through topical searches and forward/backward citation tracing. Because the aim is conceptual integration, we do not claim database exhaustiveness or report screening counts.

The analytical unit is a *claim-object-context* tuple. For each source, we ask: What object is being judged? Which assurance property is supported? At what mission stage and under what network, threat, and governance assumptions? What evidence backs the claim, and how mature is that evidence? A supplementary matrix records bibliographic identity, source type, domain, platform, mission, trust object, assurance layer, lifecycle stage, evidence, threat, trust method, blockchain role, assumptions, evaluation maturity, and limitations. Bibliographic metadata and DOI resolution were checked independently. Fine-grained method, assumption, and limitation fields are coded only after lawful full-text inspection; pending entries are explicitly marked and are not used to support detailed comparative claims. This distinction prevents a common failure in broad reviews: treating an abstract or title as if it established deployment conditions or performance.

The synthesis proceeds by conceptual comparison rather than vote counting. Convergent findings are used to define stable design pressures, such as constrained underwater links or the resource cost of on-device ledgers. Divergent findings are treated as conditional: for example, a consensus mechanism that is acceptable on shore servers may be unsuitable on a battery-limited UUV, and a trust threshold valid for routine monitoring may be unsafe for emergency intervention. Statements such as “the literature emphasizes” are attached to explicit groups of sources; the review does not infer prevalence from an incomplete corpus. The proposed OAL framework, dependency grammar, CTG loop, maturity ladder, and CDUTP are marked as the authors’ synthesis. They organize published evidence but have not been empirically validated as an integrated system.

The evidence base includes both general and domain-specific sources. General IoT and blockchain literature supplies reusable mechanisms, while cross-domain unmanned-system studies supply operational constraints. Transfer is never automatic. A trust model demonstrated in a road vehicle network, for example, may inform context-aware evidence aggregation but does not establish performance under acoustic delay; an industrial permissioned ledger may demonstrate policy consistency yet say little about disconnected underwater operation. Throughout the review, cross-domain applicability is framed as an engineering hypothesis that requires validation in the target environment.

Table 1 summarizes the boundary choices. Satellite, high-altitude, shore, cloud, and edge nodes are included as supporting infrastructure rather than additional vehicle domains. The mission lifecycle ends at audit, revocation, and recovery because trustworthy operations require action after anomalies: credentials expire, compromised devices are quarantined, policies roll back, partitions reconcile, and evidence remains available for accountability. This endpoint also exposes where blockchain has its strongest plausible value—shared state and traceable decisions across organizations—and where it has weak value—hard real-time physical control.

Table 1. Review boundary and interpretive choices.

Dimension	Included	Excluded or treated only as context
Physical domains	UAV, USV, UUV / AUV / ROV; buoys and cross-media gateways	Ground vehicles as a primary domain; platform structural certification
Infrastructure	Satellite, high-altitude, shore, cloud, edge, consortium services	Assumption that infrastructure is continuously connected or universally trusted
Mission settings	Observation, inspection, search and rescue, environmental monitoring, logistics, and publicly documented defense-support contexts	Weapon effects, targeting procedures, and sensitive operational tactics
Trust mechanisms	Identity, authorization, attestation, provenance, dynamic inference, policy, ledger, audit, governance	A new scoring algorithm, protocol proof, or claim of calibrated probability
Evidence	English peer-reviewed work, authoritative reviews, standards, official specifications	Unverified citations, inaccessible detail inferred from abstracts, and unsupported prevalence statements
Review output	Conceptual framework, taxonomy, design boundaries, maturity ladder, research propositions	New empirical data, comparative performance claims, or deployment certification

3. Operational Context and Trust Foundations

3.1. A Heterogeneous Mission System of Systems

The core system contains UAVs, USVs, and UUVs that contribute different sensing, mobility, computation, and communications functions. UAVs offer rapid repositioning, line-of-sight radio relays, and electro-optical sensing, but their payload, endurance, and link stability are constrained. USVs can carry larger energy stores and antennas, remain on station, and bridge acoustic and radio networks. UUVs provide proximity to underwater targets and environments but frequently depend on low-rate acoustic channels, short-range optical links, or delayed surfacing for high-volume transfer. Surveys of UAV-aided maritime communication and 6G maritime connectivity illustrate the relay opportunities [54,55]; underwater communication reviews document the trade-offs among acoustic, optical, and other media [56–59]. The resulting network is not a smaller terrestrial cloud. Its topology, latency, bandwidth, clock quality, and reachability can change during a single mission.

Cross-domain cooperation also changes fault containment. A USV may be the navigation reference, data cache, radio gateway, and permissioned-ledger peer for several UUVs. Compromise or benign failure of that high-betweenness platform therefore affects link evidence, time, provenance continuity, and access to the shared record. Conversely, platform diversity can improve resilience when evidence from independent media and locations can corroborate an event. Cooperation research demonstrates the control and planning value of heterogeneous formations and multi-swarms [5,6,60,61]; it does not by itself guarantee that shared observations or commands are trustworthy. Mission assurance must make such dependencies explicit.

Administrative heterogeneity is equally important. A civil emergency may combine government responders, commercial communications providers, research vehicles, and contracted data services. A coalition mission may include platforms that remain under distinct authorities. Authentication inside each organization does not automatically establish cross-organizational authority, policy equivalence, or accountability. The system boundary must therefore include organizations, credentials, delegated roles, data-use restrictions, software and model versions, and the governance process that admits, suspends, or removes participants. This is the principal condition under which a shared ledger becomes plausible: no single party should be able to rewrite the common audit state, yet all parties need a consistent basis for relying on it.

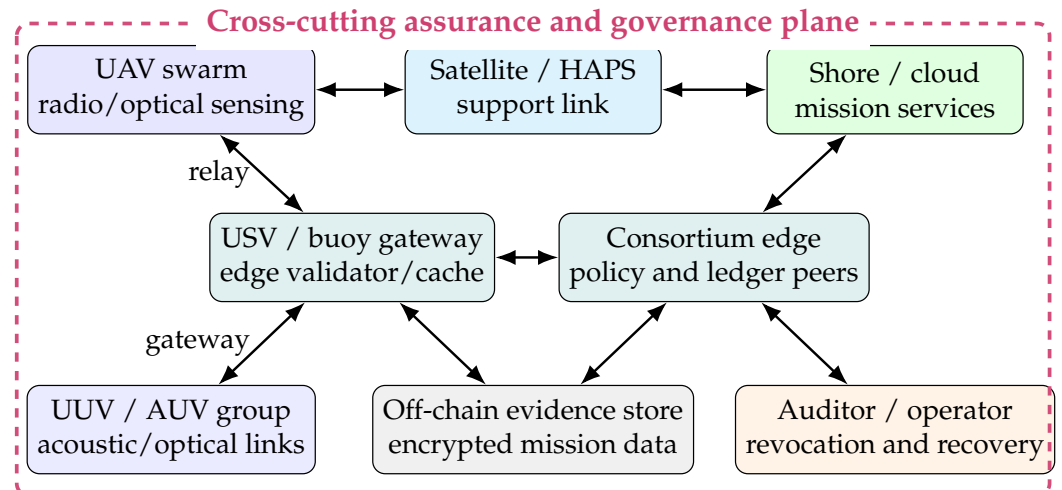


Figure 1. Author-proposed cross-domain reference architecture. Full ledger peers and policy services are placed on relatively capable USV, shore, or edge nodes; constrained vehicles retain signed local logs and submit commitments when connectivity permits. Dashed enclosure denotes the assurance and governance plane, not a universal trust boundary.

Figure 1 captures a logical deployment, not a prescribed topology. Raw imagery, sonar, telemetry, and detailed attestations remain encrypted off chain. A permissioned ledger may retain content identifiers, commitments, coarse-grained state transitions, membership changes, policy and model versions, revocations, and audit decisions. During disconnection, an edge node keeps an append-only signed sequence and later submits a batch commitment. Reconciliation can reveal gaps, duplicates, invalid credentials, or conflicting histories; it cannot retroactively satisfy a missed control deadline or prove that a sensor's physical interpretation was correct.

3.2. Mission Trust, Trustworthiness, and a Trusted Data Environment

We define *mission trust* as a context-specific relying decision about whether an object may contribute to a stated mission purpose, given available evidence and the consequences of error. Trust is therefore relational and prospective: a consumer relies on an object for an action. *Trustworthiness* denotes properties that justify such reliance, including identity authenticity, platform integrity, link availability, provenance, freshness, data quality, and accountable governance. Neither is a permanent label. The same UAV may be suitable for low-risk environmental imagery while being excluded from time-critical navigation support because its attestation is stale or its location is uncertain.

For an object o , time t , context c , and mission use m , a general representation is

$$T_o(t, c, m) = \Pr(o \text{ satisfies the properties required for } m \mid \mathcal{E}_{1:t}, c), \quad (1)$$

where $\mathcal{E}_{1:t}$ denotes the direct and indirect evidence currently available. Equation (1) is a conceptual notation. It does not assert that the probability is objectively known, calibrated, or reducible to a scalar in every implementation. Subjective logic explicitly represents uncertainty, and probabilistic or reputation systems offer useful formalisms [51–53]; a mission system may instead use sets, intervals, ordinal states, or decision rules. What matters is that uncertainty and evidence lineage are not hidden behind an apparently precise score.

A *trusted data environment* is the socio-technical setting that produces, transports, transforms, shares, and governs evidence and mission data. It includes principals, devices, paths, data products, policies, models, services, and accountability processes. Mission trust is the decision relation exercised within that environment. The distinction prevents

architectural overreach: building a tamper-evident catalog may improve the environment but does not establish that every cataloged item is fit for every decision. Similarly, zero trust is an access-control architecture, not a declaration that no component is relied upon. NIST's formulation still requires policy engines, identity and device evidence, enforcement points, and protected resources [39]; in intermittently connected missions those functions must degrade safely and make staleness visible.

3.3. Four Non-Substitutable Assurance Claims

The assurance chain in Figure 2 distinguishes four questions. Observation validity asks whether a measurement plausibly represents the physical phenomenon, considering calibration, geometry, environment, sensor health, and corroboration. Commitment and provenance integrity asks who or what produced the item, when and where it was produced, how it was transformed, and whether the recorded bytes and lineage were altered. Ledger and governance consistency asks whether authorized record keepers agree on membership, order, versions, and state transitions under stated fault assumptions. Mission-use suitability asks whether the item is timely, relevant, sufficiently certain, policy-compliant, and proportionate to the consequences of the contemplated action.

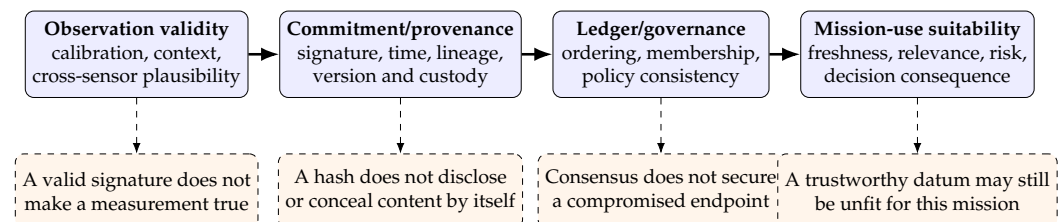


Figure 2. The authors' four-layer assurance chain. Each layer can support the next, but none substitutes for it.

These claims fail differently. GNSS spoofing can make an otherwise healthy UAV report the wrong location; machine-learning and multi-sensor approaches may detect inconsistency, but their validity depends on data and context [62–66]. A stolen signing key can produce syntactically valid provenance. A permissioned consensus group can faithfully order false submissions. A delayed but accurate observation can be unfit for collision avoidance. The chain therefore rejects four tempting equivalences: signature equals truth; hash equals privacy; consensus equals endpoint security; and historical reputation equals current mission fitness.

Remote attestation and device baselines fit between the first two layers. The RATS architecture distinguishes evidence, appraisal policy, attestation results, and relying parties [41]; the NIST IoT baseline emphasizes device identification, configuration, data protection, logical access, software update, and cybersecurity-state awareness [67]. These mechanisms can justify claims about measured software and device state. They still depend on roots of trust, reference values, verifier freshness, and the scope of measurements, and they do not certify an external physical observation. Provenance similarly records derivation rather than intrinsic truth: W3C PROV relates entities, activities, and agents, enabling consumers to ask how a product was generated [44,45]. OAL brings these distinctions into one mission-oriented vocabulary.

4. Fragmented Research Landscapes

The relevant evidence is substantial but partitioned by technical community. Table 2 compares the principal neighboring review families against the dimensions needed for mission trust. A check mark indicates sustained treatment; a circle indicates partial or enabling treatment; a dash means that the dimension is not a central organizing concern.

The comparison is interpretive rather than numerical. It shows why merely combining search terms does not close the conceptual gap: each family asks a different primary question.

Table 2. Comparison of neighboring review families and the cross-layer gap. Symbols: ✓, sustained coverage; ○, partial/enabling coverage; –, not central.

Review family	Air/sea/underwater	Four objects	Assurance chain	Dynamic uncertainty	Ledger boundary	Lifecycle	Sources
UAV networks, applications, and security	○	○	–	○	–	○	[8,14–16]
UAV blockchain	–	○	○	–	○	○	[22–25]
Unmanned marine and cross-domain cooperation	✓	–	–	–	–	○	[11,12,54,68]
Underwater communication and security	○	○	–	○	–	○	[18,21,59,69]
IoT trust and reputation	–	○	–	✓	○	○	[48–50,70]
Provenance, attestation, and zero trust	–	○	○	○	–	✓	[40,46,47,71]
General IoT blockchain	–	○	○	–	○	○	[36,37,72,73]
This review and OAL synthesis	✓	✓	✓	✓	✓	✓	Authors' proposed analytical framework

4.1. Cross-Domain Cooperation without an Assurance Vocabulary

Cooperation studies clarify what platforms can do together. They address formation control, cooperative path planning, communication support, target search, and distributed consensus in the control-theoretic sense [2–5,7,61]. These studies make heterogeneity explicit and sometimes model disturbances or link constraints. Their word *consensus*, however, usually denotes agreement of agent states or coordinated control, not replicated-ledger consensus. Security provenance, identity delegation, and cross-organizational audit are typically external assumptions. OAL treats coordination output as a data product whose inputs, transformation, policy version, and consuming mission action remain traceable.

Maritime search and unmanned-vehicle reviews show that mission effectiveness relies on distributed sensing, localization, planning, and human oversight [11,13,68]. They also demonstrate why a single platform-centered trust label is insufficient. A UUV may be healthy while its acoustic path is jammed; a USV gateway may authenticate yet relay a stale model; a UAV may provide valid imagery whose georegistration is wrong. The mission consumer needs claim-specific evidence, not a binary designation that an entire vehicle is trusted.

4.2. Security Catalogs without Cross-Object Propagation

UAV, swarm, and underwater-security reviews offer mature attack taxonomies. Typical categories include physical capture, identity and authentication attacks, GNSS spoofing, jamming, denial of service, malicious routing, eavesdropping, malware, false-data injection, and privacy leakage [14,15,17–19]. Hierarchical detection and response illustrates how monitoring and countermeasures can be distributed in a UAV network [74]. Yet an attack list does not show how an initial capability traverses objects. If a captured gateway hosts a route, transforms observations, endorses a batch, and records a commitment, defenses placed only at the ledger interface may detect neither the physical compromise nor the semantic corruption.

Environmental degradation creates an additional classification problem. Acoustic fading, multipath, Doppler, marine noise, weather, low energy, and topology change

can produce loss, latency, or disagreement that resembles attack. Underwater-network reviews repeatedly emphasize these physical constraints [56,57,75,76]. A trust system that equates anomalies with malice may isolate the very nodes needed during severe conditions. Conversely, attributing all anomalies to the environment invites an adversary to hide within expected variation. A cross-object dependency model must preserve at least three hypotheses: malicious manipulation, benign degradation, and insufficient evidence.

4.3. Trust Computation without Mission Semantics

IoT trust and reputation literature provides broad taxonomies of direct observation, recommendation, context, centralized and distributed management, and attacks on reputation [48–50,77]. Underwater studies have explored decision-tree and hidden-Markov approaches [78,79], and design guidance recognizes underwater-specific evidence constraints [80]. These methods show how a belief can be updated; they do not automatically define what the belief means for a mission. “Trust in node A” can ambiguously refer to honest routing, current software integrity, sensor accuracy, or willingness to cooperate. OAL requires each estimate to name its object, property, context, evidence lineage, uncertainty expression, and permitted use.

Blockchain-based trust management promises shared reputation and tamper-evident updates across IoT participants [70,81–83]. The ledger can reduce unilateral history rewriting, but it may also make a weak trust model more persistent. Colluding recommendations, correlated observations, model drift, and incorrect identities are upstream problems. The synthesis must consequently start from evidence semantics and only then select a storage and governance mechanism.

Fog and edge trust approaches demonstrate the attraction of placing appraisal close to constrained devices, while broader trust-recommendation reviews expose the continuing problems of context, recommendation quality, and attack resistance [84,85]. In a cross-domain mission, proximity is not synonymous with independence: a USV edge service may observe several UUVs yet remain their common gateway. A local estimator can reduce delay, but the dependency graph must record that its recommendations share a platform, clock, path, and administrative owner.

4.4. Ledger Mechanisms without Physical Truth

General blockchain reviews explain architectures, consensus mechanisms, smart contracts, scalability, and security [86–90]. IoT integrations confront limited computation, storage, communication, energy, privacy, and interoperability [36,37,91]. UAV implementations illustrate authentication, clustering, data acquisition, and swarm coordination [31,92–94]; maritime implementations show authentication and signcryption around shared logistics infrastructure [95,96]. These are useful design precedents, but their guarantees depend on membership, key custody, endorsement, validation, and the relation between on-chain commitments and off-chain content.

Cross-sector classifications show that blockchain applications inherit recurring problems of scalability, data quality, integration, and governance [97]. UAV-specific taxonomies and 6G perspectives reach a similar conclusion from the networking side: ledger mechanisms must be placed within resource, mobility, and software-defined-network constraints rather than assumed to operate uniformly on every aircraft [98–100]. These observations support selective deployment on capable gateways and edge nodes, but they do not decide which mission assurance claims deserve ledger support.

The unresolved intersection is therefore not “blockchain plus three vehicle types.” It is the absence of a common analytical model that connects a physical observation, the

responsible principals and devices, the paths and transformations, the shared governance record, and the eventual mission decision. The next section supplies that model.

5. The OAL Framework for Mission Trust

The authors' *Object–Assurance–Lifecycle* (OAL) framework is a three-dimensional classification and design instrument. It can be used to locate a published mechanism, specify a required assurance case, or identify an unprotected transition. The dimensions are intentionally orthogonal. An object says *what is judged*; an assurance layer says *which claim is sought*; and a lifecycle stage says *when the claim is needed*. A mechanism may occupy several cells, but it should not collapse the axes. Figure 3 presents the whole framework.

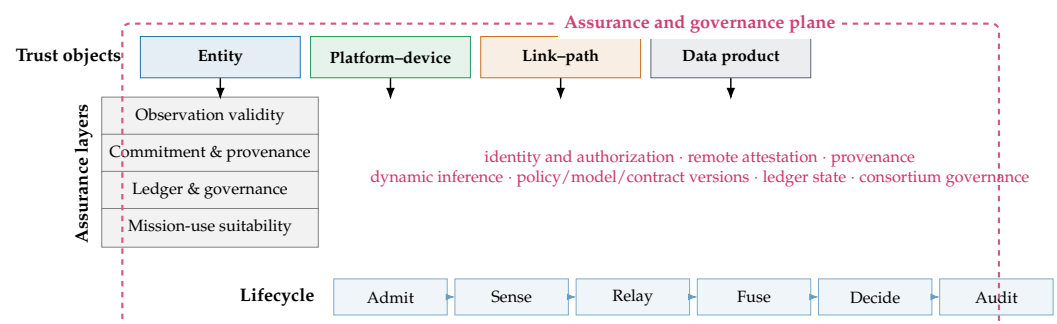


Figure 3. The authors' OAL framework. The cross-cutting plane carries mechanisms and versioned governance artifacts across all object, assurance, and lifecycle cells; it is not an additional mission object.

5.1. Dimension I: Four Trust Objects

An *entity* is a principal capable of holding authority or responsibility: an operator, organization, mission service, software agent, or autonomous decision-making principal. A *platform/device* is an executing physical or virtual endpoint: vehicle, sensor, gateway, compute module, secure element, software image, or hosted model instance. This distinction matters because an authorized entity can control a compromised device, while a device with valid attestation can execute a legitimate but unsuitable policy. An autonomous vehicle may occupy both categories under different claims: as a delegated principal for action and as an endpoint whose hardware and software state are measured.

A *link/path* is a time-varying channel or end-to-end route, including its media, intermediate nodes, security associations, and quality state. It is not reducible to a physical radio or acoustic hop. A *data product* may traverse a UUV acoustic hop, a USV gateway, a satellite backhaul, and a cloud broker; confidentiality, freshness, availability, and routing integrity depend on the composed path. A *data product* is an immutable version of raw observation, feature set, track, map, fused estimate, model output, command, alert, provenance record, or audit artifact. Treating versions as immutable avoids silent overwriting and permits provenance to express transformation.

Table 3. Four OAL trust objects, evidence, failure examples, and mission questions.

Object	Representative evidence	Illustrative failure or attack	Mission question
Entity	Credential chain, organizational role, delegation, authorization, prior accountable actions	Credential theft, insider misuse, collusion, obsolete authority	Who requests, endorses, or consumes the resource, and under which authority?
Platform/device	Secure boot measurement, firmware/software version, sensor self-test, configuration, energy and health state	Capture, malware, unsafe configuration, stale reference value, sensor drift	Which endpoint executes, and is its measured state adequate for this use?
Link/path	Authentication state, route, delay, loss, signal indicators, neighbor changes, medium and gateway state	Jamming, wormhole, selective forwarding, eavesdropping, benign fading	Through which changing path did evidence pass, and what security and quality properties held?
Data product	Signature, timestamp, location, provenance graph, quality meta-data, version, calibration and cross-source residuals	False injection, replay, poisoning, misregistration, unauthorized transformation	Where did this version come from, was it altered, and is it timely and suitable?

Policies, models, smart contracts, and ledger state do not create a fifth object category. They are versioned artifacts hosted on devices or represented as data products, while the principals who authorize them remain entities. Keeping four operational objects prevents an ontology from expanding whenever a new mechanism is introduced. At the same time, the assurance plane makes these artifacts first-class subjects of versioning, access control, threat analysis, and audit.

5.2. Dimension II: Four Assurance Layers

The first layer, *observation validity*, asks whether evidence about the external world is credible. Relevant controls include calibration records, sensor diagnostics, geometry, physical bounds, environmental models, independent modalities, and temporal continuity. Multi-view and multi-sensor fusion can expose disagreement and express uncertainty [101–103]; it can also amplify correlated error or poisoned inputs. Observation assurance must document independence assumptions and cannot be replaced by a content hash.

The second layer, *commitment and provenance integrity*, binds an item to a producer, time, place, transformation, and immutable version. Digital signatures, authenticated channels, secure timestamps, content identifiers, and W3C PROV relations support this layer [44–47,104]. Provenance should cover model inference and human or automated decisions, not only raw acquisition. Decision provenance can connect inputs, policy versions, intermediate transformations, and outputs, making later review possible. The guarantee remains conditional on key custody, clock assumptions, instrumentation completeness, and trustworthy capture of events.

The third layer, *ledger and governance consistency*, concerns a shared state among administrative parties. It covers member admission, endorsement, record ordering, validation, version coordination, revocation, dispute handling, and reconciliation. Hyperledger Fabric illustrates an execute–order–validate architecture with identity-aware channels and endorsement policies [105,106]; other permissioned designs can satisfy similar functions. This layer is about authorized agreement under a declared fault model. It does not add semantic truth to endorsed data.

The fourth layer, *mission-use suitability*, converts assurance evidence into a bounded relying decision. Freshness, relevance, uncertainty, risk class, legal and organizational policy, available alternatives, and consequence severity are considered. A consumer may accept a low-resolution image for broad situational awareness but require corroborated, current, and accurately georegistered evidence for autonomous maneuver. Suitability must

therefore be expressed as a purpose-specific authorization or confidence condition, not a universal trust badge.

5.3. Dimension III: Six Lifecycle Stages

The lifecycle dimension prevents security from ending at admission. Table 4 shows how objects and assurance questions evolve from credentialing to recovery. Evidence can age, dependencies can change, and an initially valid decision can require reassessment.

Table 4. Mission lifecycle and representative OAL assurance tasks.

Stage	Primary questions	Evidence and controls	Ledger-relevant state
Admission and authentication	Is the entity authorized, and is the joining endpoint in an acceptable measured state?	Credentials, delegation, attestation, configuration baseline, least privilege	Membership, role, attestation-result commitment, policy version
Sensing and collection	Does an observation plausibly reflect its stated time, place, modality, and calibration?	Sensor self-test, secure time/location, quality metadata, corroboration	Acquisition commitment and data identifier
Transmission and relay	Which path carried the item, and did quality or security conditions change?	Mutual authentication, route and link telemetry, delay/loss context, custody transfer	Signed relay events or batch roots
Processing, fusion, and sharing	Which inputs, code, model, parameters, and policies produced this version?	Provenance graph, reproducible transformation identifier, access and usage control	Version, endorsement, access decision, model/policy commitment
Decision and action	Is the product fit for the specific mission consequence now?	Uncertainty, freshness, risk policy, alternatives, human review or fail-safe	Decision rationale commitment and responsible principal
Audit, revocation, and recovery	What failed, who was affected, what must be invalidated, and how is operation restored?	Incident evidence, credential and model revocation, rollback, re-attestation, after-action review	Revocation state, dispute outcome, reconciled log, recovery version

Lifecycle continuity is especially difficult during partitions. If a UUV cannot contact a central policy engine, it still needs a local authorization envelope: named resources, permitted actions, risk limit, evidence freshness, expiry, and fail-safe behavior. Local actions are signed and sequenced; reconnecting nodes later disclose commitments and selected evidence. The design must distinguish *continued operation under delegated authority* from *fresh global authorization*. A ledger can make the difference auditable but cannot remove it.

5.4. The Assurance and Governance Plane

The cross-cutting plane contains identity and authorization, remote attestation, provenance capture, trust inference, policy/model/contract version management, ledger state, and consortium governance. Each component is itself attackable. Identity authorities can be compromised; reference values can be wrong; provenance instrumentation can omit transformations; a trust model can drift or be poisoned; a smart contract can encode an unsafe rule; validators can collude; and governance can deadlock. The plane must therefore be observed through the same OAL lens. For example, a model is a data product with provenance and validation evidence, a deployed instance is hosted on a device, its approval is an entity action, and its use is constrained by lifecycle stage and mission context.

OAL is useful as both a diagnostic and a specification template. A study that reports blockchain throughput but leaves the observation, endpoint, and mission layers unspecified occupies a narrow set of cells. An attestation protocol that measures firmware but does not

address disconnected freshness covers a device claim at admission but not necessarily at decision time. A mature mission-assurance design should identify every high-consequence dependency path, state the claim at each cell, and expose which evidence is absent or stale.

6. Evidence Dependencies, Threat Propagation, and Dynamic Reasoning

6.1. A Typed Object-Dependency Graph

OAL classifies claims; a dependency graph explains how they interact. Let $G_t = (V_t, E_t)$ be a time-indexed directed multigraph whose vertices are versioned instances of the four object types. Edges carry typed relations: an entity *controls* a platform; a platform *hosts* a sensor, model, or gateway; a platform *observes* a data product; a link *transmits* it; an activity *transforms* inputs into a new product; an entity or service *endorses* it; a ledger transaction *records* a commitment; and a mission function *consumes* it. Time and version qualifiers are essential. “UAV-7 hosts model M” is insufficient if the decision depended on model version 4.2 before a rollback.

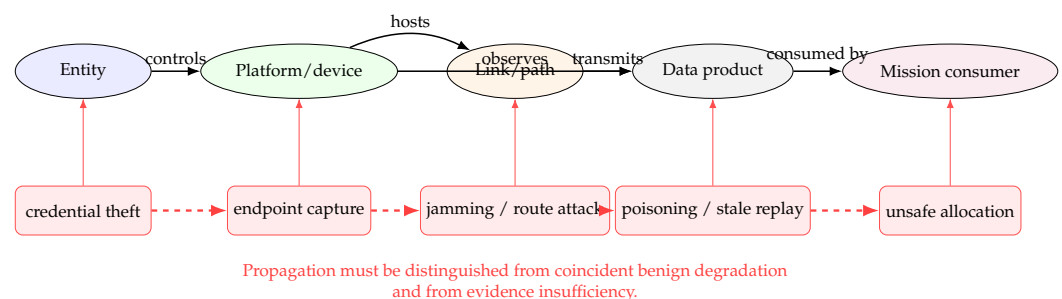


Figure 4. Simplified object dependency and attack-propagation path. Red dashed arrows describe a possible causal path, not proof that every anomaly is malicious.

The graph unifies evidence and threat reasoning. Evidence can support a local claim and propagate conditionally: a fresh attestation supports the integrity of the platform that generated a product; verified provenance supports the relation between an input and transformation; independent observations support physical plausibility. An attack can traverse the same relations in the opposite epistemic direction. Stolen entity credentials authorize access to a device, device compromise corrupts observations or gateway functions, a manipulated path delays or substitutes a product, and a mission consumer acts on the result. Figure 4 gives one illustrative chain. The graph does not imply simple multiplication of trust scores. Dependencies can be correlated, defensive evidence can be incomplete, and path effects can be non-monotonic.

6.2. A Unified Threat-Analysis Sentence

Every scenario should be expressible as: *an attacker with capability C acts on object O, propagates through relations P, creates mission impact I, and is constrained or detected by controls K under assumptions A.* This grammar connects threat catalogs to assurance cases. It also exposes missing controls. Table 5 applies the grammar to common public scenarios without specifying tactical exploitation procedures.

Table 5. Object-centered threat, degradation, and evidence-absence analysis.

Class and initial object	Capability or condition	Propagation path	Mission impact	Controls and residual uncertainty
Malicious/entity	Stolen or misused credential	controls device → endorses product → record accepted	Unauthorized tasking or false accountability	Short-lived delegation, multi-party approval, behavioral evidence, rapid revocation; valid credentials can still mask insider intent
Malicious/device	Capture, malware, or model replacement	hosts sensor/model → observes/transforms data	Consistent but false products across many consumers	Attestation, key protection, diversity, provenance, quarantine; measured components may omit sensors or runtime behavior
Malicious/link	Jamming, wormhole, selective forwarding, replay	transmits/custodies data → delays or substitutes version	Loss of availability, stale situational picture, biased fusion	Link context, authenticated sequence, path diversity, freshness policy; attribution remains uncertain
Malicious/data	False-data injection, poisoning, semantic manipulation	transformed / endorsed / recorded product → consumed decision	Incorrect classification, route, or resource allocation	Physical plausibility, independent modality, robust fusion, lineage; independence assumptions can fail
Benign environment	Fading, multipath, weather, occlusion, low energy	degrades link or observation evidence without hostile action	Delay, disagreement, reduced coverage	Environment-aware baselines, graceful degradation, explicit “benign-degraded” hypothesis; models may be incomplete
Evidence absence	Disconnection, failed verifier, missing lineage, stale clock	prevents appraisal across several relations	Unsafe overconfidence or unnecessary isolation	Unknown state, bounded authority, conservative policy, later reconciliation; absence is not proof of compromise

Three-way interpretation is a minimum, not a full classifier. *Malicious* means that evidence supports intentional or adversarial manipulation. *Benign-degraded* means that an environmental or operational explanation is supported. *Unknown* means evidence cannot justify either conclusion. A recovery state is useful after remediation because newly attested software does not instantly restore confidence in cached data, keys, neighbors, or prior decisions. Policies should map these states to bounded actions—continue, reduce privilege, request corroboration, isolate, return, or escalate to a human—rather than convert them into moral labels for entire platforms.

6.3. The CTG Reasoning Loop

We use CTG-Trust only as a name for the authors’ context–temporal–graph reasoning loop shown in Figure 5. It is not presented as a novel estimator. First, evidence events are captured with source, subject, time, context, value, and lineage. Second, evidence reliability is appraised: a device self-report and an independent verifier do not carry the same evidential relationship. Third, object state is inferred using an appropriate published method. Fourth, dependencies are traversed to identify downstream claims and shared causes. Fifth, uncertainty is made explicit. Sixth, a mission policy chooses an action. Seventh, the decision, relevant versions, and evidence commitment are recorded for accountable review. Eighth, operational outcomes and later verification update the evidence base.

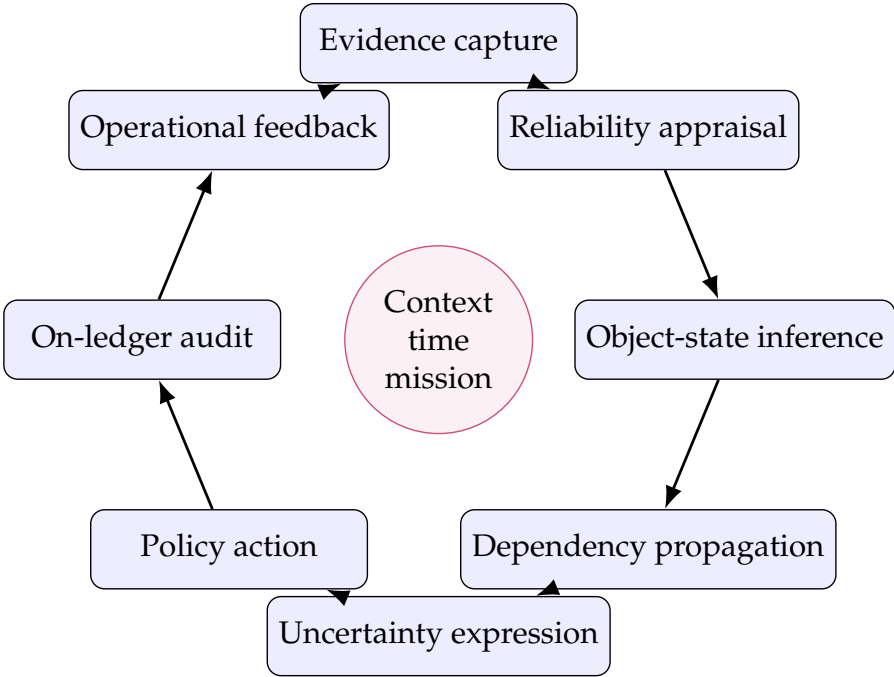


Figure 5. CTG as an author-proposed reasoning loop. The figure defines information flow and accountability, not an unvalidated scoring algorithm.

Evidence reuse requires special care. A gateway’s report can influence device reputa-
tion, link assessment, data quality, and a final fused decision; treating each derived value
as independent counts the same evidence repeatedly. The dependency graph should retain
parent identifiers and common causes so that an implementation can discount correlation or
avoid duplicate use. Recommendations also require provenance. A high reputation value
from ten pseudonyms is not ten independent observations, and ledger persistence does not
remove a Sybil or collusion problem unless membership and organizational independence
are enforced.

6.4. Trust Methods and Appropriate Claims

The literature offers several method families, summarized in Table 6. None is univer-
sally superior because each expresses a different claim under different assumptions. The
comparison is qualitative and should guide future validation, not be read as benchmark
evidence.

Table 6. Trust-inference families and their role in CTG.

Family		Useful representation	Strength	Critical assumption/risk	Representative sources
Weighted attribute score	multi-	Context-weighted evidence features	Interpretable and implementable at constrained edges	Weights and thresholds can be arbitrary or task-insensitive	[48,49]
Bayesian/reputation		Evidence-updated belief or behavior propensity	Natural sequential updating	Priors, stationarity, and evidence independence may be unjustified	[52,53]
Subjective logic		Belief, disbelief, uncertainty, and base rate	Makes ignorance visible and supports opinion combination	Dependence and source identity must be handled explicitly	[51]
State-space/Markov		Latent state with temporal transitions	Represents persistence and recovery	State meanings and transition parameters need mission data	[79]
Rule/tree/fuzzy methods		Human-readable rules or learned partitions	Can combine heterogeneous evidence	Coverage gaps and abrupt boundaries; calibration often unclear	[78,80]
Machine/deep learning		Learned anomaly or multi-view representation	Captures nonlinear patterns	Shift, poisoning, explainability, and overconfidence	[62,63,102]
Graph-based propagation	propa-	Relational evidence over dependencies	Connects local evidence to downstream exposure	Correlation, cycles, and collusion can amplify error	[53,81]

Equation (1) can sit above any of these methods if its meaning remains modest. A score must state the event whose probability or degree it represents, its context, horizon, calibration evidence, and decision loss. Without such information, two decimal places create precision without assurance. For this reason the framework does not specify Beta parameters, hidden-state transitions, scoring thresholds, message schemas, or performance targets. Those are empirical design choices for future studies.

7. Blockchain-Assisted Assurance and Its Limits

7.1. The Decision to Use a Ledger

Blockchain is justified by a governance problem before it is justified by a data problem. A permissioned ledger is plausible when multiple independently administered parties must coordinate shared state, no single party should control history, records require accountable endorsement, and intermittent participants must reconcile after disconnection. A conventional authenticated database is usually preferable when one trusted organization controls membership and audit, when a hard real-time loop cannot wait for distributed confirmation, or when there is no meaningful cross-organizational dispute. This test narrows the design space and avoids adding consensus overhead merely because unmanned nodes are distributed.

The relevant ledger is normally permissioned. Open, computation-intensive participation is poorly aligned with constrained vehicles, sensitive mission metadata, and accountable membership. Full peers and ordering services belong on capable shore, edge, or selected USV nodes; UAVs and UUVs can act as clients or light verifiers. Hyperledger Fabric provides one representative architecture with identities, endorsement policies, channels, ordering, and validation [105,106]. The OAL framework is ledger-neutral: Byzantine-fault-tolerant replicas, crash-fault-tolerant ordering, or another shared-state service may be selected if the stated threat, availability, and governance assumptions match. Consensus surveys show that throughput, finality, membership, energy, and fault tolerance involve trade-offs rather than a single ranking [87,88,107].

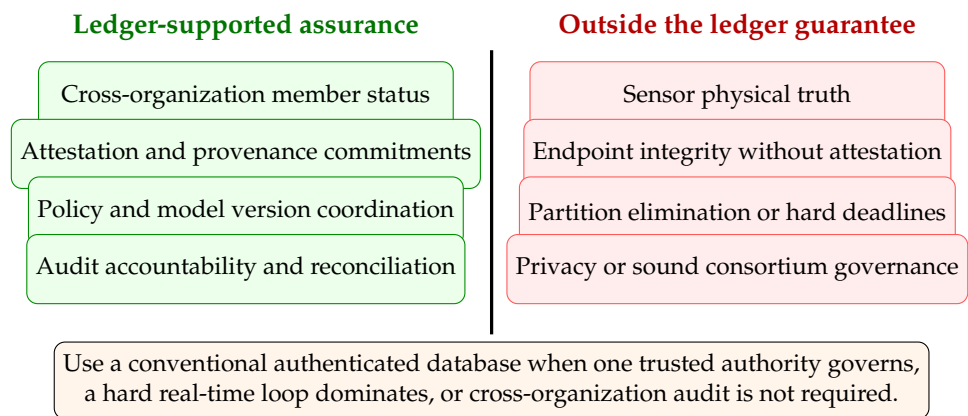


Figure 6. Blockchain boundary in mission trust. The left column lists plausible support; the right column lists claims a ledger does not establish on its own.

7.2. On-Chain and Off-Chain Partitioning

Raw video, sonar, point clouds, high-rate telemetry, detailed attestation evidence, and fine-grained location should remain off chain, encrypted and subject to data-use policy. The ledger records small, verifiable references: content identifiers or Merkle roots, producer and custodian identifiers, coarse timestamps or time ranges, schema and policy versions, attestation-result commitments, endorsement decisions, revocations, and audit outcomes. The relation is bidirectional. An off-chain object must be verifiably bound to its commitment, and a consumer must be able to determine which ledger state and authorization applied when it was created and used.

On-chain hashes are integrity commitments, not privacy mechanisms. A hash can leak equality, support dictionary attacks on predictable values, and bind sensitive metadata to a durable public-to-members record. Encryption, access control, selective disclosure, retention, key destruction, aggregation, and metadata minimization remain necessary. Similarly, deleting an off-chain object does not erase its on-chain metadata. Privacy design must account for traffic analysis, membership visibility, timing, and the linkage of vehicle identifiers across missions. General IoT and blockchain reviews identify privacy and storage tension [35,36,108]; maritime governance adds commercial and jurisdictional concerns [32,33,109].

7.3. Four Design Patterns

Table 7 organizes recurring ledger uses by OAL claim, assumptions, and limits. These patterns are the authors’ synthesis of UAV, IoT, provenance, and attestation work rather than standardized solutions.

Table 7. Blockchain design patterns for cross-domain mission assurance.

Pattern	On-chain record	Assurance supported	Necessary assumptions	What remains unresolved
Membership and authority registry	Organization, role, credential status, delegation and revocation version	Cross-party identity and policy-state consistency	Governed issuer set, protected keys, bounded offline credentials	Real intent, endpoint integrity, emergency authority conflict
Attestation anchor	Evidence digest, verifier identity, result, reference-value and policy version	Accountable device-state appraisal and freshness history	Trustworthy measurement root and verifier; adequate coverage	Sensor truth, unmeasured runtime behavior, compromised verifier
Provenance commitment	Data/version identifier, transformation and custody roots, endorsement	Tamper-evident lineage and accountable processing	Complete instrumentation, secure time and keys, retrievable off-chain evidence	Semantic correctness, biased models, confidentiality of metadata
Partition-aware audit and reconciliation	Signed sequence range, batch root, local policy/model version, reconnect event	Detection of omission, duplication, conflicting logs, and stale authority	Secure local append state, retained evidence, eventual reachability	Missed real-time deadline, events never captured, physical truth
Policy/model coordination	Approved artifact identifier, activation/rollback time, authorized endorsers	Shared knowledge of active rule or model version	Clear governance and deterministic validation of identifiers	Safety and performance of the approved artifact

The membership pattern generalizes UAV identity and authentication proposals [27,28,31]. Its key question is not whether a public key appears on a chain, but which organization may issue which role, under which policy version, and how offline parties learn of revocation. Short-lived credentials, bounded delegation, and risk-specific offline envelopes reduce but do not eliminate the stale-revocation problem.

The attestation pattern links RATS evidence appraisal to shared audit. An attester produces evidence; a verifier evaluates it under an appraisal policy; a relying party uses a result [41]. Recording a commitment to the result and policy can make later reliance accountable and can coordinate verifier outputs among organizations. Attestation architectures for blockchain and edge systems illustrate this intersection [42,43,110,111]. The ledger should not store sensitive measurement detail by default, and the result's validity interval must reflect disconnection and device state change.

The provenance pattern links W3C-compatible derivation to shared commitments. IoT provenance proposals use blockchain to protect provenance records or bind large off-chain collections [112–115]. For mission data, each derived product should identify immutable inputs, transformation code or model, parameters, responsible entity, and output version. Provenance supports explanation and accountability, including for AI artifacts [116]; it does not make a biased transformation correct. Consumers still appraise observation validity and mission suitability.

The partition-aware pattern is especially relevant to underwater missions. A disconnected node or gateway keeps a signed, monotonically sequenced local log, periodically commits a batch root, and discloses inclusion evidence after reconnecting. Consortium peers validate membership and authority at event time, sequence continuity, duplicates, and policy/model versions. Competing logs can be flagged for governance. This resembles store-carry-forward accountability more than online transaction processing. The CAP trade-off is inescapable: under a partition, a system cannot simultaneously provide unrestricted availability and immediate global consistency [117]. The mission must specify which local actions remain available and how uncertainty is represented.

7.4. Governance, Revocation, and Recovery

Permissioned consensus relocates trust; it does not remove it. A consortium must define member eligibility, validator distribution, endorsement thresholds, emergency powers, software upgrade, key rotation, data jurisdiction, dispute resolution, incident disclosure, and exit. If one operator controls all validators and governance, the ledger provides at most internal tamper resistance and engineering convenience; it does not create multi-party

independence. Conversely, requiring every organization to endorse every event can destroy availability. Governance policy must be threat- and mission-specific.

Revocation is temporal. A credential revoked at 12:00 may have authorized a disconnected UUV at 11:55; a gateway may learn the change at 12:30. Audit needs event time, receipt time, local policy version, credential validity evidence, and reconciliation time. Recovery also extends beyond identities. Compromised models, data products, and provenance descendants may require invalidation; devices require re-attestation; mission decisions may require human review. A ledger can preserve the dependency identifiers needed to compute the affected set, while the OAL graph explains why a simple credential-state flip is insufficient.

7.5. Real-Time and Safety Boundaries

Hard control loops should not wait for ledger finality. Vehicle stabilization, collision avoidance, emergency surfacing, and other deadline-bound safety functions require local verified control and deterministic timing. A ledger can distribute approved policy or software versions before operation and record summaries afterward. It may support slower supervisory decisions, cross-party authorization, data release, and after-action audit. This separation also protects availability: loss of the ledger service should degrade audit freshness and cross-organizational coordination, not disable fundamental platform safety.

Figure 6 is therefore a claim discipline. Blockchain may strengthen who agreed to what, under which version, and in which order. It cannot alone establish that the world matched the record, that every endpoint was secure, that a network stayed connected, that a deadline was met, or that the consortium's rules were legitimate. Any paper that claims more must supply independent physical, endpoint, timing, privacy, and governance evidence.

8. Evaluation Practices and Evidence Maturity

The literature spans conceptual architectures, analytical models, simulations, prototypes, datasets, and a smaller number of domain demonstrations. These forms answer different questions. A security analysis can expose protocol assumptions but not acoustic-channel realism; a network simulation can explore scale but not organizational governance; a blockchain prototype can measure transaction latency while accepting fabricated sensor inputs; a field trial can demonstrate integration at modest scale without testing adversarial behavior. Comparing claims without identifying their evidence level produces false equivalence.

8.1. An Evidence-Maturity Ladder

Figure 7 proposes a six-level ladder for claims about cross-domain mission trust. It is an ordering of ecological evidence, not a single quality score. Strong mathematical analysis at level 2 can be more informative about a security property than a poorly controlled field demonstration at level 4. Advancement means that additional classes of assumption are exposed and tested. A claim should name the level actually reached and avoid language belonging to a higher level.

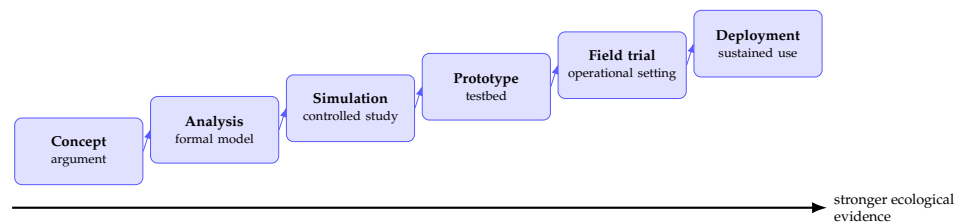


Figure 7. Authors' evidence-maturity ladder. Progress requires preserving lower-level clarity while adding implementation, environmental, organizational, and longitudinal evidence.

At level 0, a work offers a use case, taxonomy, or architecture with traceable reasoning. Many blockchain opportunity papers properly occupy this level; their value is conceptual, not proof of feasibility. Level 1 adds mathematical, formal, or security analysis with explicit adversary and system assumptions. Level 2 uses simulation or recorded data to test behavior under controlled variation. Level 3 integrates an implementation in a prototype or testbed and reports configurations, code, and workload. Level 4 conducts trials in representative air, surface, or underwater conditions with realistic communications and operational participants. Level 5 demonstrates sustained multi-party operation, including governance changes, credential and software rotation, failures, audits, and recovery. A ledger throughput test cannot alone move a complete mission-trust claim beyond level 3, because the observation and mission-use layers remain untested.

8.2. Datasets, Simulators, and Testbeds

No public artifact identified here directly covers UAV-USV-UUV motion, heterogeneous links, device attestation, provenance, adaptive trust, consortium governance, and mission outcomes in one benchmark. Existing resources can nevertheless form a modular validation stack. CICIOT2023, TON_IoT, and Edge-IIoTset offer large-scale or heterogeneous IoT attack telemetry [118–120]; they do not encode ocean physics or vehicle dependencies. SeaDronesSee provides real maritime UAV imagery and metadata for detection tasks [121]; it is not a cybersecurity dataset. GNSS spoofing studies provide real or realistic signals and detection methods [64,65], but they represent one attack and platform context.

AirSim supports high-fidelity visual and physical simulation for autonomous vehicles [122]. UUV Simulator integrates underwater robots with Gazebo/ROS [123], while AquaSim NG provides ns-3-based underwater network modeling [124]. General ns-3 design work illustrates the value of open, composable network simulation, although any acoustic, radio, and mobility model must be validated for the intended scenario. These tools can be coupled through timestamped event interfaces, but coupling is itself an experimental contribution: time synchronization, shared geometry, link abstraction, and reproducibility need documentation.

Table 8. Evaluation resources, the OAL cells they can exercise, and transfer limits.

Resource/practice	Primary evidence	OAL use	Important metrics	Transfer limitation
CICIoT2023, TON_IoT, Edge-IIoTset	Network, host, sensor, and attack telemetry	Device/link anomaly and uncertainty methods	Detection by class, calibration, false isolation, delay, missingness sensitivity	Generic IoT topology; no cross-media maritime mission
SeaDronesSee and GNSS records	Maritime imagery, metadata, navigation anomalies	Observation validity, data-product quality, spoofing context	Detection error, geolocation error, uncertainty calibration, robustness to weather/viewpoint	Limited security labels; not a ledger or underwater benchmark
AirSim	UAV/vehicle physics, sensors, controllable scenarios	Air-domain observation and mission use	Mission success, safety constraint violations, sensor/model shift	Does not natively reproduce acoustic networks or consortium governance
UUV Simulator	ROS/Gazebo underwater vehicles and intervention	Underwater motion, sensing, control, integration	Localization, energy, mission completion, failure recovery	Simulated hydrodynamics and sensors require field validation
Aqua-Sim NG/ns-3	Underwater-network protocols and channels	Link/path evidence, delay, loss, routing and partition	Delivery, latency distribution, energy, path churn, outage duration	Network abstraction does not establish observation truth
Permissioned-ledger testbed	Peers, ordering, endorsement, contracts, failures	Ledger/governance consistency and audit	Commit/finality latency, throughput, resource use, recovery, fork/conflict handling	Synthetic transactions may ignore real evidence sizes and duty cycles
Cross-domain field exercise	Physical platforms, operators, weather, real links	Integrated lifecycle and mission-use suitability	Consequence-weighted mission outcomes, false quarantine, recovery time, audit completeness	Costly and scenario-specific; adversarial coverage and repeatability limited

8.3. Metrics that Respect the Assurance Chain

Evaluation should report at least five metric families. First, *inference quality*: discrimination where labels exist, calibration or interval coverage, uncertainty under missing evidence, detection and recovery delay, and performance under distribution shift. Accuracy alone is inadequate because a rare false isolation can remove a critical relay. Second, *mission effect*: task completion, time, coverage, safety-constraint violations, decision regret, and resource allocation under adversarial and benign degradation. Third, *network cost*: bytes per evidence event, delay distribution rather than only mean delay, energy, storage growth, duty cycle, and partition duration. Fourth, *ledger behavior*: endorsement and finality latency, throughput under realistic event batching, validator/resource cost, membership change, outage behavior, reconciliation completeness, and disputed histories. Fifth, *accountability*: percentage of decisions with resolvable provenance, time to determine affected descendants, policy/model version traceability, and successful revocation and recovery.

Experimental factors should cross object, threat, and context dimensions. Examples include compromised entities versus devices; isolated versus colluding nodes; jamming versus environmental fading; fresh versus stale attestation; independent versus correlated sensors; honest versus compromised gateways; short versus prolonged partitions; and routine versus high-consequence mission policies. A factorial design can reveal interactions that single-attack demonstrations miss. Crucially, ground truth must distinguish malicious action, benign degradation, and missing evidence. Otherwise a detector can appear successful by learning weather or topology artifacts.

Baselines should also be layered. A centralized authenticated database tests whether a blockchain adds value; a signed local log tests whether replication is needed; static access control tests the value of continuous evidence; object-local trust tests graph propagation; a no-provenance condition tests the value of lineage; and a no-uncertainty condition reveals overconfident decisions. Blockchain comparisons should hold endorsement, workload, hardware, and fault assumptions constant. Studies of Fabric-based access control and integrated IoT ledgers offer implementation starting points [112,125]; they do not remove the need for an application-specific baseline.

Reproducibility requires the exact topology, channel and mobility models, workload, cryptographic settings, ledger release and configuration, endorsement and fault model, random seeds, data partitions, trust-policy versions, and mission-loss function. Versioned provenance should cover the evaluation itself. This is more than archival hygiene: if a

policy threshold or model changes between runs, the reported mission outcome belongs to a different data product and assurance claim.

8.4. Reading Current Claims Conservatively

UAV blockchain applications range from security and privacy architectures to swarm monitoring and resilient-network proposals [94,126–129]. Their reported evidence should be read at the level actually provided. Likewise, broader claims about blockchain in Industry 4.0, federated learning, digital twins, supply chains, and government sharing demonstrate reusable organizational patterns [130–134]. Transfer to intermittent, safety-relevant cross-domain missions remains a hypothesis until endpoint, link, timing, governance, and mission consequences are jointly tested.

The supplementary matrix is designed to support this discipline. It separates a DOI-verified source record from a full-text-coded claim and leaves unresolved detail visible. Before submission, all sources supporting method-specific comparisons or strong gap statements should be checked in full text, with paywalled items supplied lawfully by the authors. This draft intentionally avoids inventing maturity ratings or performance values for unavailable texts.

9. Discussion and Research Agenda

9.1. Theoretical Implications

OAL reframes trust from a property attached to a vehicle into a structured relying relation. This has three theoretical consequences. First, trust is compositional but not freely transitive. Confidence in an entity can support a claim about authorization, and device attestation can support a claim about execution state, yet neither automatically transfers to an observation or mission decision. Composition requires typed dependency relations and explicit inference rules. Second, trust is purpose-bound. Evidence sufficient for sharing low-sensitivity imagery may be insufficient for cooperative navigation, even when the underlying object is identical. Third, uncertainty is operational state. “Unknown” should produce a documented policy response, not be silently mapped to trusted or malicious.

The assurance chain also clarifies the relationship between provenance and truth. Provenance is an argument about history: who generated an artifact, through which activities, from which inputs. Observation validity is an argument about the world. Ledger consistency is an argument about shared records. Mission suitability is an argument about consequence. These arguments can reinforce each other but remain falsifiable independently. Provenance and decision-accountability research supports the value of traceable transformations [104,116]; OAL adds the requirement that a mission consumer name which history is sufficient for which use.

9.2. System Resilience under Partition and Degradation

Future architectures should treat partition as an expected mode, not an exceptional outage. Local policy envelopes require bounded capability, expiry, evidence-age limits, safe fallback, and explicit record of which global facts were unavailable. Reconnection then becomes a semantic reconciliation problem, not merely data synchronization. The system must decide whether actions were authorized at event time, whether a newly received revocation invalidates descendants, and whether conflicting observations require retrospective mission review. Blockchain scalability research focuses on throughput and storage [135,136]; cross-domain missions add long disconnection, sparse contacts, and consequence-sensitive local autonomy.

Resilience also requires functional diversity. If the USV is simultaneously acoustic gateway, time source, fusion node, policy-enforcement point, and validator, redundant

vehicle count can conceal a single logical point of failure. The dependency graph can expose this concentration by measuring which vertices support many mission paths. Future work should compare architectural redundancy (multiple instances) with evidential independence (different roots, media, sensors, and organizations). Two reports relayed through the same compromised gateway are not two independent confirmations.

9.3. Governance, Privacy, and Human Authority

Consortium governance is part of the security mechanism. Research should specify who can admit validators, approve reference values, activate contracts and models, invoke emergency powers, decide disputes, and audit the auditors. Maritime blockchain adoption studies already identify policy and organizational factors beyond technical performance [32,34,109]. In mission assurance, a governance failure can propagate like a cyberattack: an unsafe reference value makes compromised devices appear healthy, or a delayed revocation keeps stale authority active.

Privacy questions extend beyond encrypting payloads. Cross-mission identifiers, validator endorsements, timing, route custody, and model-use records can reveal organizational relationships or vehicle activity. Work on blockchain and IoT warns of immutable metadata and access-control tension [36,72,137]. Future designs should evaluate unlinkability, data minimization, selective disclosure, authorization-policy confidentiality, and deletion semantics alongside audit value. A mission may need accountable pseudonymity: an operator can verify that an authorized role acted without learning the long-term identity, while a governed process can resolve accountability after an incident.

Human authority must remain explicit. Automated trust inference can recommend reduced privilege, corroboration, or isolation, but high-consequence recovery and dispute decisions may require human review. The relevant evidence includes not only model output but explanation, policy version, uncertainty, alternatives, and consequence. Human override itself must be authorized and auditable without turning operators into a presumed trusted endpoint.

9.4. Testable Research Propositions

The framework yields propositions that future studies can falsify.

P1 (assurance separation). Systems that expose separate observation, provenance, ledger, and mission-suitability states will reduce unsafe acceptance of correctly signed but semantically invalid data compared with systems using a single node-reputation score, under matched false-isolation cost.

P2 (dependency awareness). Trust decisions that preserve evidence lineage and shared-cause dependencies will be better calibrated under correlated sensors and colluding recommenders than decisions that aggregate object scores independently.

P3 (three-way diagnosis). Policies that distinguish malicious, benign-degraded, and unknown states will achieve lower mission loss during mixed environmental and adversarial conditions than binary trusted/untrusted policies.

P4 (bounded offline authority). Time-, resource-, and consequence-bounded offline authorization will maintain higher mission availability than fail-closed global authorization while producing lower retrospective policy violation than unrestricted offline autonomy.

P5 (selective ledger value). A permissioned ledger will improve cross-organizational audit completeness and dispute resolution relative to signed centralized logs when governance is genuinely distributed, but will not improve hard real-time control performance and may degrade it if placed in the control path.

P6 (maturity gap). Designs evaluated only through transaction or network metrics will overstate mission assurance relative to evaluations that combine observation validity,

endpoint compromise, path degradation, governance change, and consequence-weighted mission outcomes.

Each proposition requires operational definitions and preregistered decision losses in future empirical work. They are not findings of this review. Their purpose is to convert broad research gaps into comparative designs with refutable outcomes.

9.5. From OAL Cells to an Assurance Case

OAL becomes operational when each high-consequence mission use is expressed as a compact assurance case. The top claim should be deliberately narrow: for example, “fused product d_{17} is suitable for route replanning by consumer c until time t under weather class w and policy p .” The argument then descends through the four layers. Observation subclaims identify the sensors, calibration, geometry, environmental model, residuals, and corroboration that justify physical plausibility. Commitment subclaims identify immutable inputs, producers, transformations, custody, time, and protected versions. Ledger subclaims state the membership, endorsement, ordering, and policy state shared by organizations. Mission subclaims connect freshness and uncertainty to an explicit consequence and permitted action. Any unsupported subclaim is displayed as a defeater, caveat, or reason to restrict action.

This structure should include negative claims. A verifier may establish that a measured software component matches an approved reference but explicitly exclude camera alignment and operator intent. A provenance record may establish derivation while excluding completeness because a legacy transformation is not instrumented. A ledger may establish that three authorized peers endorsed a model version while excluding its safety or accuracy. Writing exclusions alongside guarantees is more informative than a long control list, because it prevents downstream consumers from silently broadening the claim.

Assurance cases must also be time-aware. Evidence has an observation time, issue time, receipt time, and expiry condition; a dependency can change between them. If a USV attestation is fresh when it receives a UUV batch but stale when it transforms and endorses that batch, the graph should not let the earlier result support the later activity without an explicit continuity assumption. Clock uncertainty must be represented where GNSS is unavailable or suspected. For disconnected operation, the case records the last globally known policy and the local authorization envelope, rather than retroactively pretending global agreement.

Counter-evidence is first-class. A new sensor residual, revocation, incident report, or conflicting custody event should identify which claims and descendants become disputed. This enables proportional response: quarantine a derived track without necessarily rejecting every observation from a platform, or reduce a gateway’s endorsement role while retaining its emergency relay function. Recovery closes the argument only after affected credentials, software, models, data descendants, and mission decisions are reviewed. A simple “trusted again” flag is insufficient.

Future tools could generate these arguments from the typed dependency graph, but automation must not hide judgment. Policy authors still decide evidence sufficiency, independence, consequence, and acceptable uncertainty. The generated artifact should therefore distinguish machine-verified bindings (signature, hash, sequence), verifier appraisals (measured state against reference), statistical assessments (calibration or anomaly), and governance judgments (waiver, dispute, emergency authority). A reviewer can then ask whether the conclusion exceeds the strongest supported subclaim.

Finally, evaluation should measure the assurance case itself. Completeness can be assessed as the fraction of consequence-relevant dependencies with current evidence; soundness can be probed through seeded counterexamples; usability can be measured by

the time and error with which operators identify affected products; and governance quality can be tested through conflicting but authorized decisions. These measures complement detection and ledger performance. They ask whether the architecture supports defensible reliance, which is the conceptual goal of mission trust.

9.6. *Toward a Cross-Domain Unmanned Trust Profile*

The proposed Cross-Domain Unmanned Trust Profile (CDUTP) would be an interoperability profile rather than a new universal protocol. It would define a small, versioned set of claims and evidence references for the OAL cells most often crossed by organizations. NIST Zero Trust Architecture would supply resource- and request-centered policy principles [39]; IETF RATS would supply roles and semantics for attestation evidence, appraisal, and results [41]; W3C PROV-DM and PROV-O would supply entity-activity-agent provenance relations [44,45]. NIST's IoT device baseline would inform minimum endpoint capabilities [67]. 3GPP specifications for UAS support and maritime services would provide communications and service context [138,139]. Permissioned-ledger governance and endorsement would supply multi-party state coordination [105,106].

Software and model update evidence should be compatible with secure-update architecture rather than invented as ledger-specific metadata. The IETF SUI architecture separates update manifests, authorization, payload acquisition, installation, and device constraints [140]; CDUTP could reference such an update identifier and result while keeping the firmware payload off chain. This illustrates the profile's intended role: bind existing domain claims into a mission assurance graph, not duplicate their protocols.

A profile instance should minimally state: object identifier and type; claim and assurance layer; issuer and relying role; evidence or provenance reference; observed, issued, and expiry times; mission context and permitted use; uncertainty or appraisal outcome; applicable policy, model, schema, and contract versions; privacy label and disclosure conditions; parent dependencies; revocation reference; and on/off-chain binding. It should support disconnected issuance and later verification without pretending that a stale result is current. Compact encodings are desirable for acoustic links, but semantic equivalence is more important than one wire format.

Standardization should proceed in stages. First, publish use cases and negative requirements, especially claims the profile must not imply. Second, define a shared vocabulary and conformance tests for evidence lineage and version binding. Third, run plugfests across independently implemented UAV, USV, UUV, gateway, verifier, and ledger components. Fourth, evaluate privacy and governance with multiple organizations, including member exit and disputed recovery. Fifth, pursue alignment with relevant standards bodies. CDUTP remains the authors' research agenda; neither its name nor the proposed fields are endorsed by NIST, IETF, W3C, 3GPP, Hyperledger, or any other organization.

9.7. *Design Heuristics*

Several practical heuristics follow. Start with the relying decision, not the ledger. Name the object and assurance claim before choosing a trust method. Preserve evidence identity and common causes. Keep raw evidence off chain and minimize metadata. Make staleness and unknown state visible. Separate hard real-time safety from shared audit. Treat policies, models, contracts, and reference values as versioned, threat-exposed artifacts. Test a centralized signed-log baseline. Finally, match language to maturity: "proposed," "analyzed," "simulated," "prototyped," "field-tested," and "deployed" are not interchangeable.

These heuristics also explain when blockchain should be rejected. If a single accountable manager already controls state and audit, signed append-only logs and replicated databases may be simpler. If no independent parties need a common history, consensus

adds little. If a platform cannot maintain the keys, time, or protected local state needed to bind evidence, a ledger commitment merely preserves unreliable input. And if a decision deadline is shorter than end-to-end communication, local safety assurance must dominate.

Responsible deployment adds one final boundary. Mission-trust infrastructure can improve safety, environmental observation, inspection, disaster response, and coalition accountability, but the same persistent identities and provenance traces can enable excessive surveillance or expose sensitive operational patterns. An assurance case should therefore include a legitimate-purpose claim, proportional data collection, retention and disclosure limits, and an accountable path for human appeal or incident review. Dual-use assessment should focus on capabilities and governance rather than assume that a defensive label removes risk. Public reporting can disclose architecture, assurance semantics, failure modes, and evaluation methods while withholding credentials, exact operational configurations, exploitable topology details, and mission-specific tactics. This balance makes results scientifically testable without converting the review into operational guidance for harmful use. It also reinforces the core OAL insight: governance artifacts and privacy policies are versioned, contestable evidence within the mission system, not external promises that can be taken on trust.

10. Limitations of the Review and Framework

This narrative synthesis is intentionally selective and therefore susceptible to author judgment in source discovery, grouping, and emphasis. It uses English-language evidence and a principal 2016–2026 window, so relevant earlier work and publications in other languages may be underrepresented. The topic spans control, robotics, communications, cybersecurity, distributed systems, provenance, trust modeling, and governance; vocabulary differs enough that no finite query guarantees complete coverage. The evidence matrix improves traceability but does not turn the corpus into an exhaustive sample or justify prevalence estimates.

Full-text access is incomplete at the draft stage. Bibliographic metadata, DOI resolution, and source routing have been checked, but method-level assumptions and limitations are retained as pending where a lawful full text has not yet been inspected. Such sources support orientation only, not detailed performance or comparative claims. Before submission, the authors should supply or obtain lawful copies of priority paywalled papers, complete the pending coding fields, verify every strong synthesis statement against the matrix, and remove or soften claims that cannot be supported. The present manuscript contains no invented result intended to fill those gaps.

Much evidence is transferred from adjacent domains. UAV security, IoT reputation, road-vehicle trust, industrial provenance, and permissioned enterprise ledgers provide mechanisms, but their threat distributions, time scales, connectivity, energy, and governance may differ from air–surface–underwater operations. Even within maritime systems, acoustic conditions and operational practices vary widely. The OAL framework makes transfer assumptions visible; it does not validate them.

OAL, the dependency relation set, CTG loop, maturity ladder, propositions, and CDUTP are the authors' conceptual synthesis. They have not been evaluated as an integrated architecture, calibrated with mission data, formally proven, or standardized. The four object types may require refinement for human teams, biological sensors, digital twins, or legal jurisdictions. The graph can become large and cyclic, and practical implementations will need rules for abstraction, correlation, privacy, and evidence expiry. The general notation $T_o(t, c, m)$ should not be interpreted as a calibrated probability without empirical validation.

Finally, public sources cannot fully represent military governance, contested environments, or sensitive platform behavior. The review deliberately excludes weapon effects and sensitive tactics. This limits operational specificity but preserves a defensible, dual-use focus on assurance, resilience, audit, and responsible governance.

11. Conclusions

Air–surface–underwater cooperation turns trust into an end-to-end mission problem. A defensible decision depends on more than an authenticated vehicle or an immutable record: it requires a plausible observation, an intact and accountable provenance chain, a correctly governed shared state where one is needed, and evidence that remains suitable for the particular mission consequence. The authors’ OAL framework connects these four assurance layers to entity, platform/device, link/path, and data-product objects over the complete lifecycle from admission to recovery.

The object-dependency graph and CTG loop show how evidence, uncertainty, and attacks move across that structure. They also impose an important restraint: environmental degradation and missing evidence must not be labeled malicious without support, and a precise trust score must not be presented as calibrated unless it has been validated. The evidence-maturity ladder extends the same restraint to published claims.

Blockchain has a necessary but limited role in settings where independent organizations need consistent membership, attestation and provenance commitments, version coordination, accountable audit, and post-partition reconciliation. It cannot establish sensor truth, protect a compromised endpoint, eliminate partitions, meet hard real-time deadlines, provide privacy through hashing alone, or substitute for consortium governance. Where one trusted manager is sufficient or no shared audit dispute exists, a conventional authenticated system may be the better design.

The proposed CDUTP and six research propositions provide a route from conceptual synthesis to interoperable, testable systems. Future work should validate the framework with lawful full-text evidence, cross-domain testbeds, representative field trials, and multi-party governance exercises. The central standard of success is not whether a ledger reaches consensus, but whether every consequential mission use can be traced to appropriate, current, and honestly bounded assurance.

Supplementary Materials: The following supporting information accompanies the submission: Table S1, the internal evidence matrix (CSV); Table S2, the lawful full-text acquisition manifest (CSV); and the editable vector source for Figures 1–7 (TikZ/LaTeX).

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Abbreviations

AUV	Autonomous underwater vehicle
CDUTP	Cross-Domain Unmanned Trust Profile
CTG	Context-temporal-graph
DLT	Distributed ledger technology
OAL	Object-Assurance-Lifecycle
RATS	Remote Attestation procedureS
UAS	Unmanned aerial system
UAV	Unmanned aerial vehicle
USV	Unmanned surface vehicle
UUV	Unmanned underwater vehicle
ZTA	Zero trust architecture

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